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AI Smart Workstation for Industrial Quality Control

Enhancing productivity through vision systems, real-time assistance, and Axiomatic Design

***Wissenschaftliche Gesellschaft für Arbeits- und
Betriebsorganisation Forschungsseminar 2025,
25-27/09/2025, TU Hamburg, Hamburg***

Greetings

A few words about myself

Education:

MSc in Data Science

University of Trento, Italy (2020-2023)

MSc in Cognitive Neuroscience

University of Trento, Italy (2016-2018)

BSc in Cognitive Science

University of Trento, Italy (2013-2016)

Work Experience:

Data Scientist

Fraunhofer Italia IEC, Bolzano, Italy (Feb 2024-Ongoing)

Data Science Researcher (Thesis)

Fondazione Bruno Kessler, Trento, Italy (Mar 2023- Mar 2024)

Data Scientist

AlmaViva SpA, Rovereto, Italy (Jan 2023-Jan 2024)

Data Analyst and Software Developer

Edizioni Centro Studi Erickson, Trento, Italy (Jan 2021-Jan 2023)



The research team – Fraunhofer Italia Research IEC



Leonardo Venturoso

MSc Data Science
Research Associate at
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Dieter Steiner

Dipl.-Ing. (M.Sc.)
Mechatronics Engineering
Team Leader at Process
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Simone Garbin

MSc Computer Science
Research Associate at
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Intelligence Development



Dominik T. Matt

Prof. Dr.-Ing.
Director at
Fraunhofer Italia

Activities at M.IN.D – Machine Intelligence Development team

Fraunhofer Italia Research – IEC

Industrial Projects on:

A) Artificial Intelligence and Computer Vision for Quality Control



B) Optimization for Flexible Production Systems and Logistics



Motivation

Why a new approach for Quality Control is needed?

Background:

Modern manufacturing is shifting toward **high-mix, low-volume production**.

Traditional quality control struggles:

1. Manual → inconsistent, Automated → too rigid.
2. **Reflective/transparent materials → defects hard to detect.**
3. **Industries** like premium packaging and automotive **demand higher customization and stricter quality standards.**



Image source: itemit.com

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Need: **scalable, modular, cost-efficient solution**.

Potential research gap

Regarding Quality Control:

Industry 4.0 offers **AI, cyber-physical systems (CPPS), smart workstations** as a starting point.

RGB (color, texture) and **polarization imaging** (surface stress, micro-defects) are two of the main strategies for imaging but are rarely integrated.

Synchronization + real-time processing = major challenge.

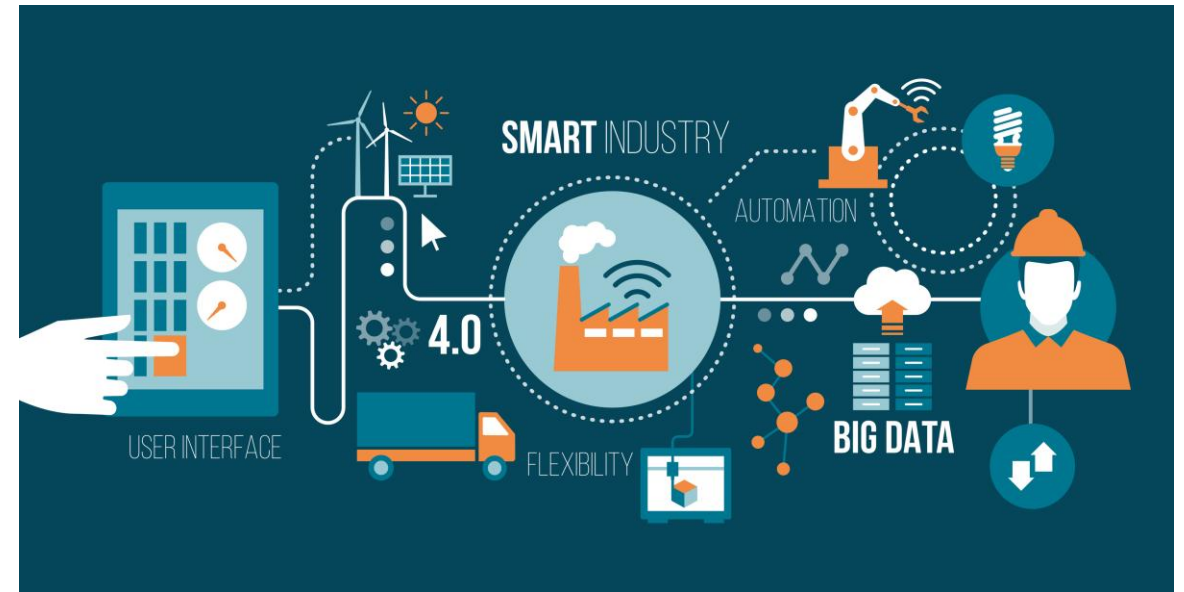


Image source: elteterni.it

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Research question: **Can RGB + PolarSens imaging improve QC in real-time?**

Axiomatic Design (AD) Framework

A useful structured framework:

Links Customer Needs (CNs) → Functional Requirements (FRs) → Design Parameters (DPs)

Benefits: Ensures **minimal complexity, modularity, scalability for SMEs, traceability.**

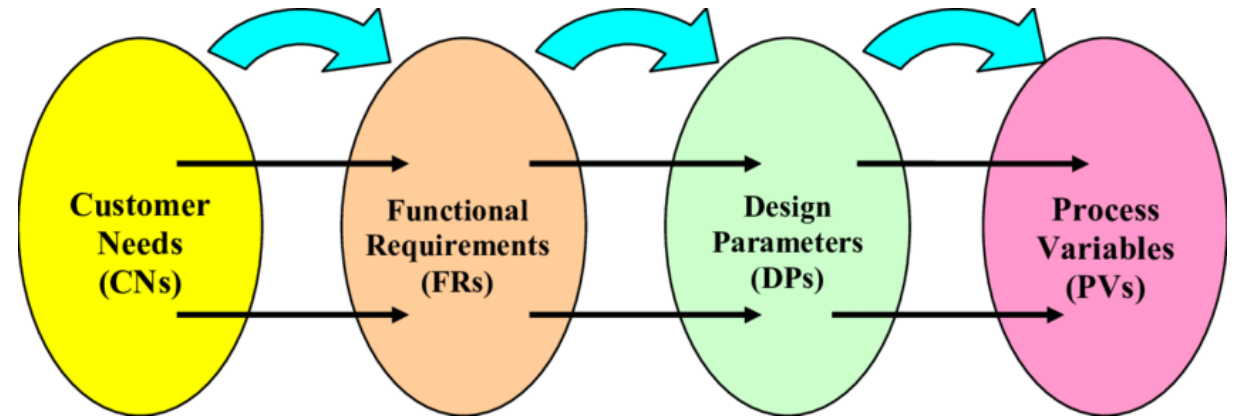


Image source: [George Draghici's Lab](#)

Industrial use case: Shower Cabin Factory

Context:

- Pre-packaging quality control for shower cabin components.
- Production characterized by high-mix, low-volume batches.

Challenges:

Manual inspection → time-consuming, inconsistent, error-prone.

Reflective and transparent materials (glass, aluminum) → difficult defect detection.

QR code labels and profiles must be verified before packaging.



Image source: Fraunhofer Italia

Industrial use case: Shower Cabin Factory

Opportunity:

Develop a modular workstation, following AD principles, to:

1. **Automate repetitive inspection** tasks.
2. **Ensure reliable detection** of defects and labels.
3. **Provide operator support** through real-time assistance.



Image source: Fraunhofer Italia

Industrial use case: Shower Cabin Factory

Axiomatic Design Approach

	DP1	DP2	DP3	DP4	DP5
FR1	X				
FR2		X			
FR3			X		
FR4				X	
FR5					X

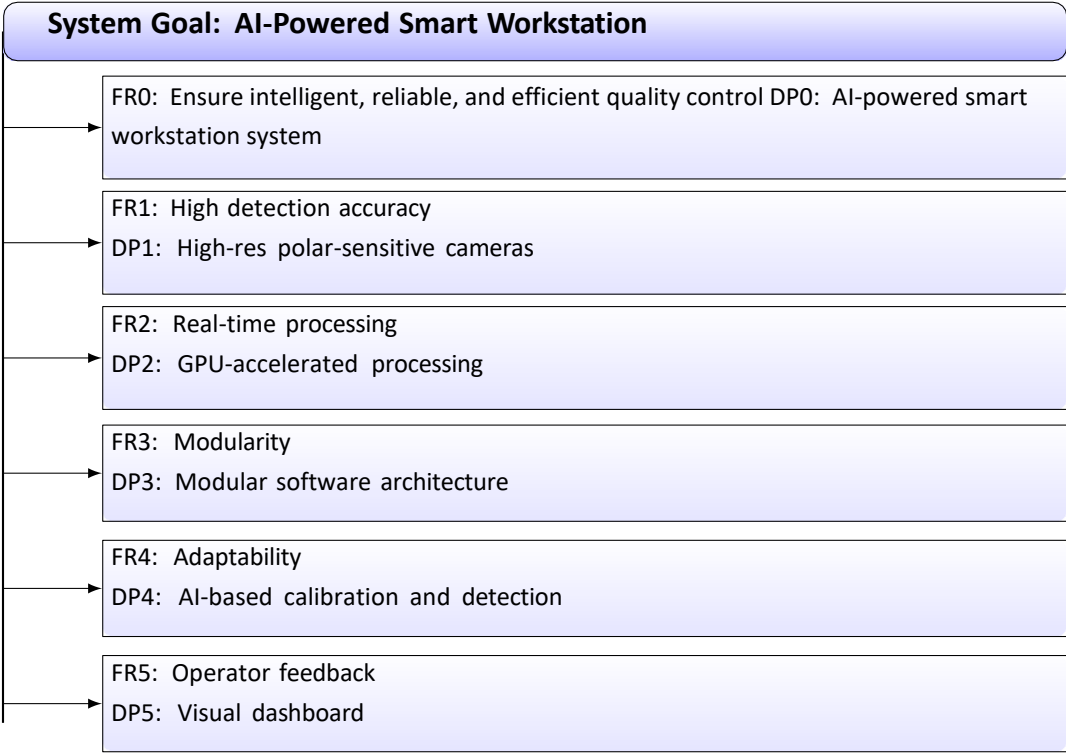


Image source: Fraunhofer Italia

Smart Workstation for Quality Control

Towards practical solutions

A CPPS integrating hardware, software, and human interaction.

Hardware:

- **RGB high-resolution camera** for QR codes and material classification.
- **Polarization camera** for reflective/transparent surface inspection.
- **Adaptive lighting system** with softboxes + tunable LEDs.

Software:

- **YOLOv11** deep learning for defect detection and recognition.
- **Modular Python pipeline** with GPU acceleration.

Human-Machine Interface:

Real-time digital assistance via Flask & web-based dashboard.

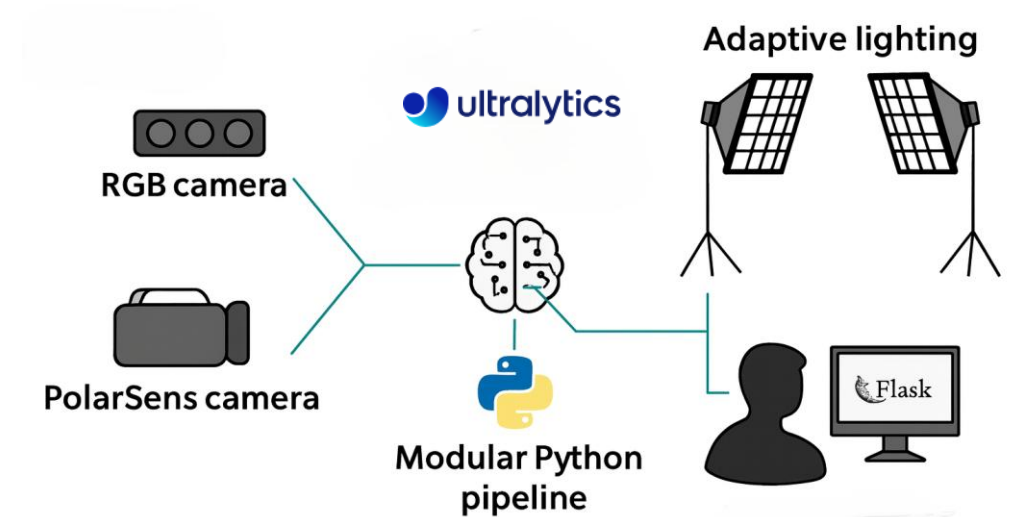


Image source: Fraunhofer Italia

Prototype Setup:

Experimental Configuration:

- RGB Camera (**Baumer VLT-X-650 C.I**) at 2.5 m height (client's need) → QR codes + recognition.
- PolarSens Camera (**Baumer VCXG.2-51MP**) → surface defect detection.
- GPU (**NVIDIA GTX 5070**) for real-time inference (~5 fps).
- Lighting system ensures glare-free, consistent imaging.

All integrated through a **plugin-ready software** pipeline.

Operator interacts via an **intuitive dashboard** with live feedback.

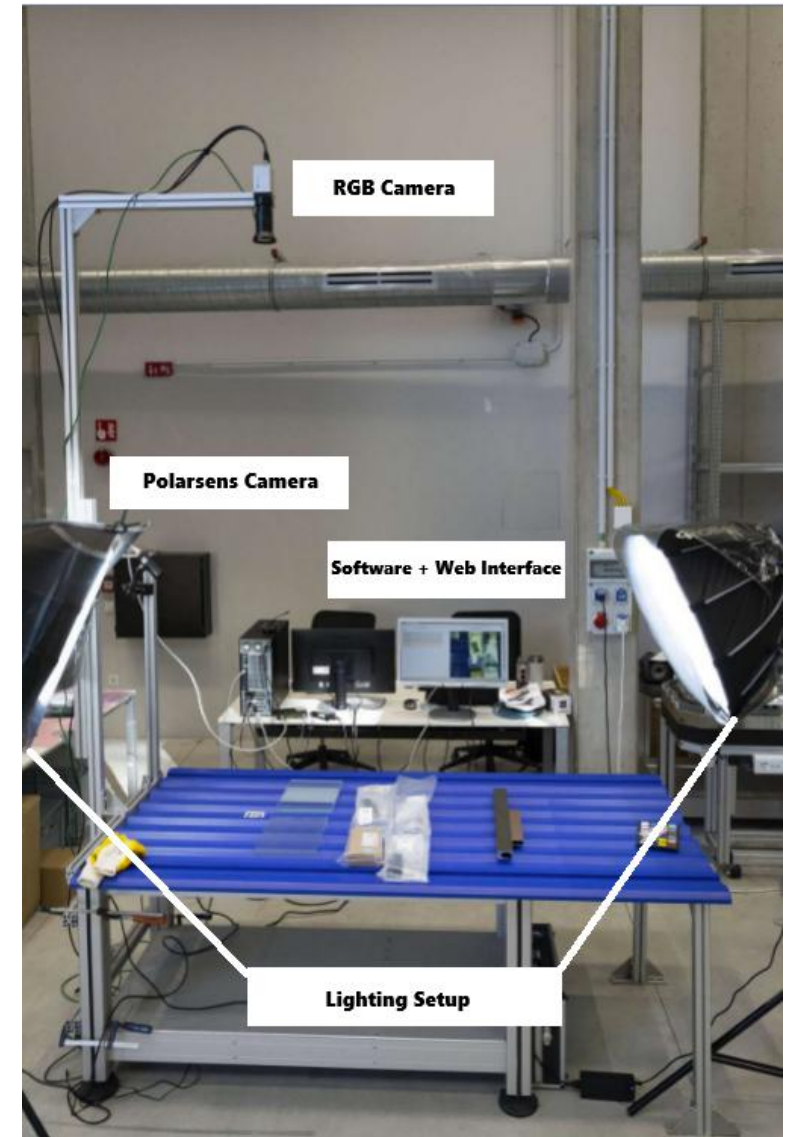


Image source: Fraunhofer Italia

Performance Results

Detection Accuracy:

>80% (QR codes, profiles, surface defects).

QR Decoding:

>80% accuracy across lighting conditions.



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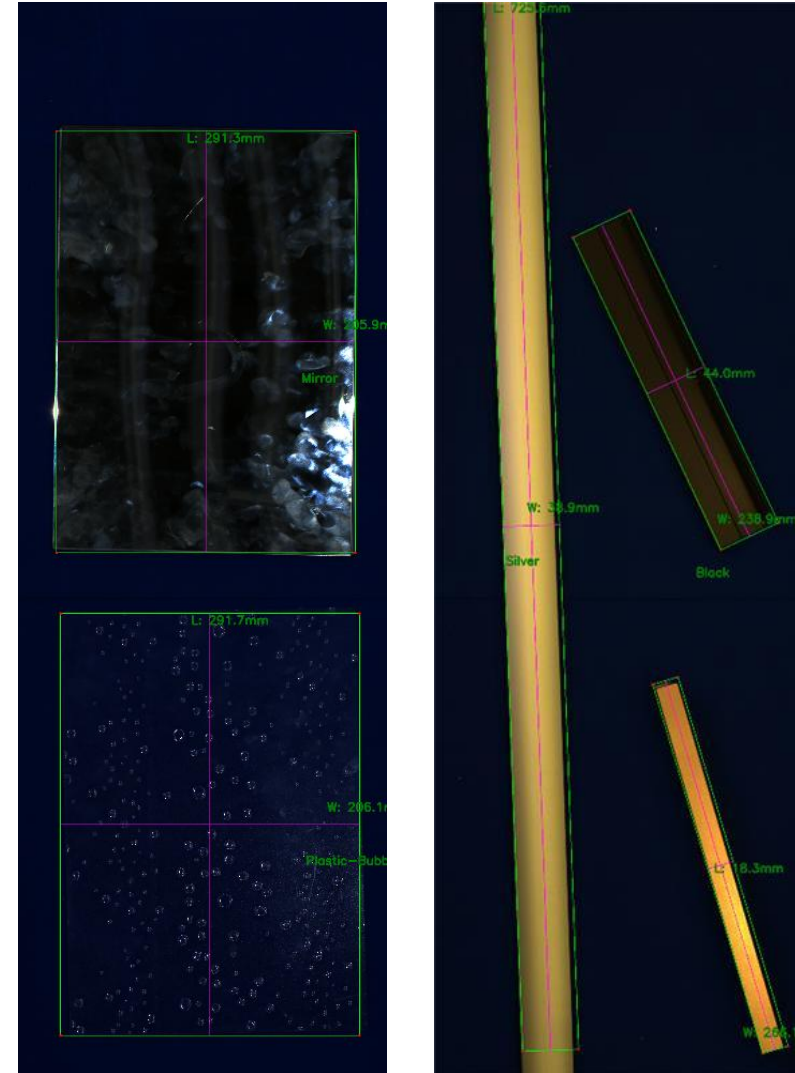


Image source: Fraunhofer Italia

Performance Results

Material Recognition:

>90% mAP50, ~70% mAP50:95.

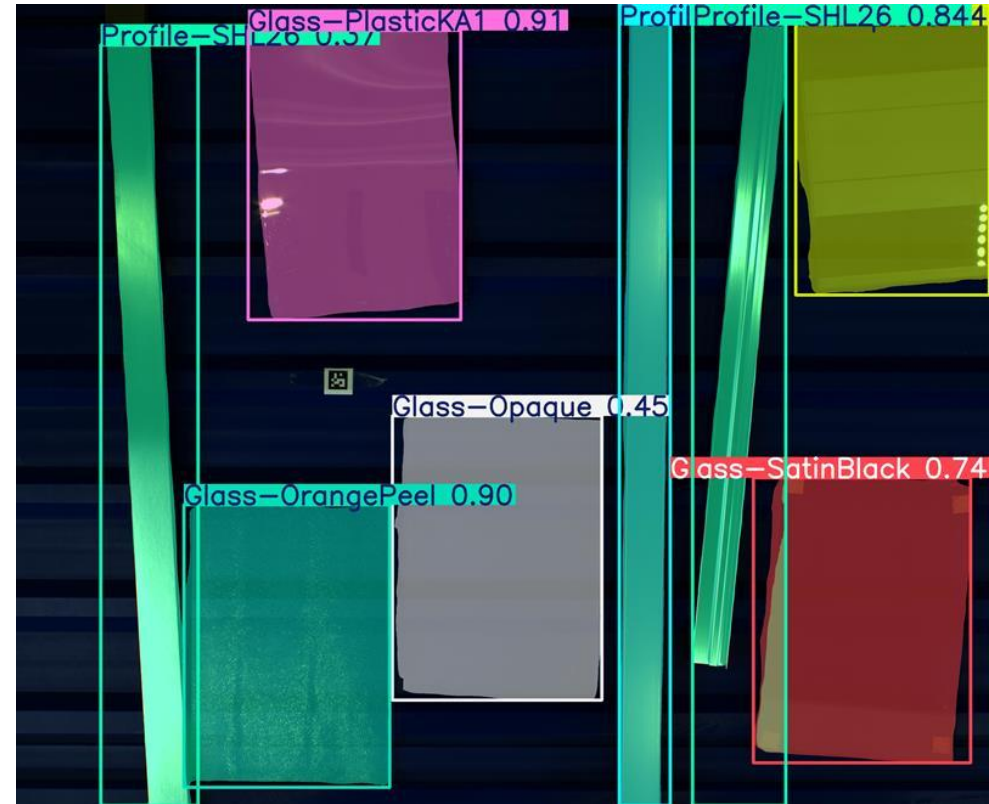


Image source: Fraunhofer Italia

Performance Results

Defect Detection:

~50% mAP50, ~20%
mAP50:95 (good baseline
for complex reflective
surfaces).

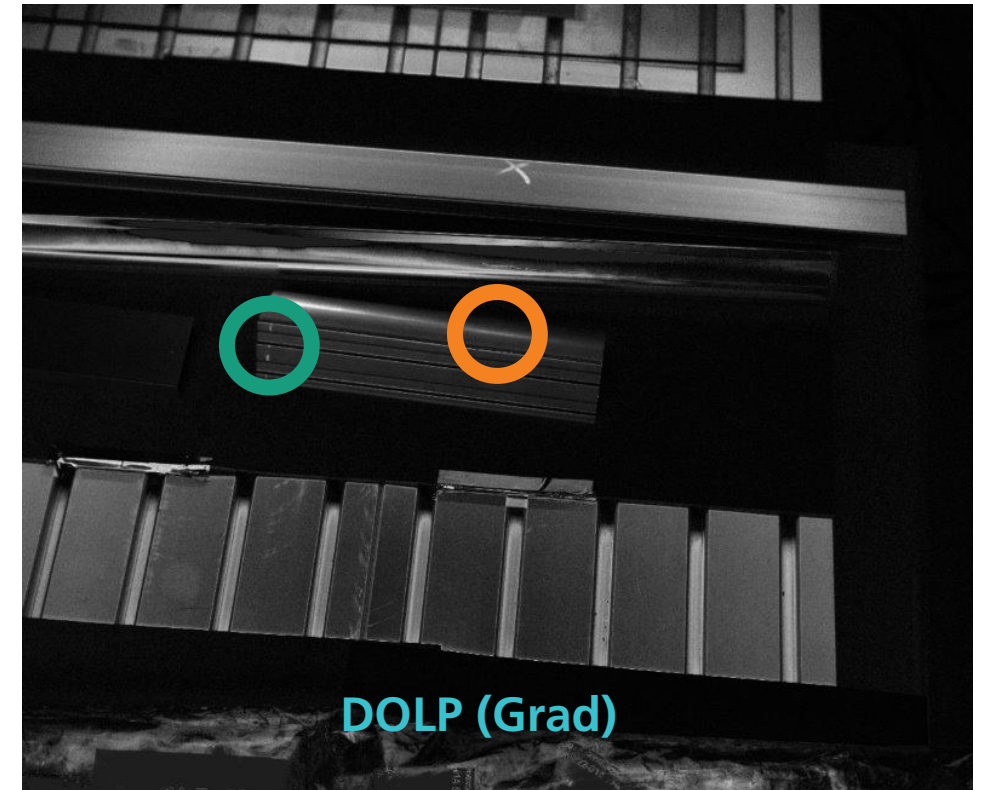


Image source: Fraunhofer Italia

Performance Results

Defect Detection:

~50% mAP50, ~20% mAP50:95 (good baseline for complex reflective surfaces).

Processing Time: 0.19 s/frame (≈ 5 fps) $\rightarrow$ 7.5 $\times$ faster than CPU.

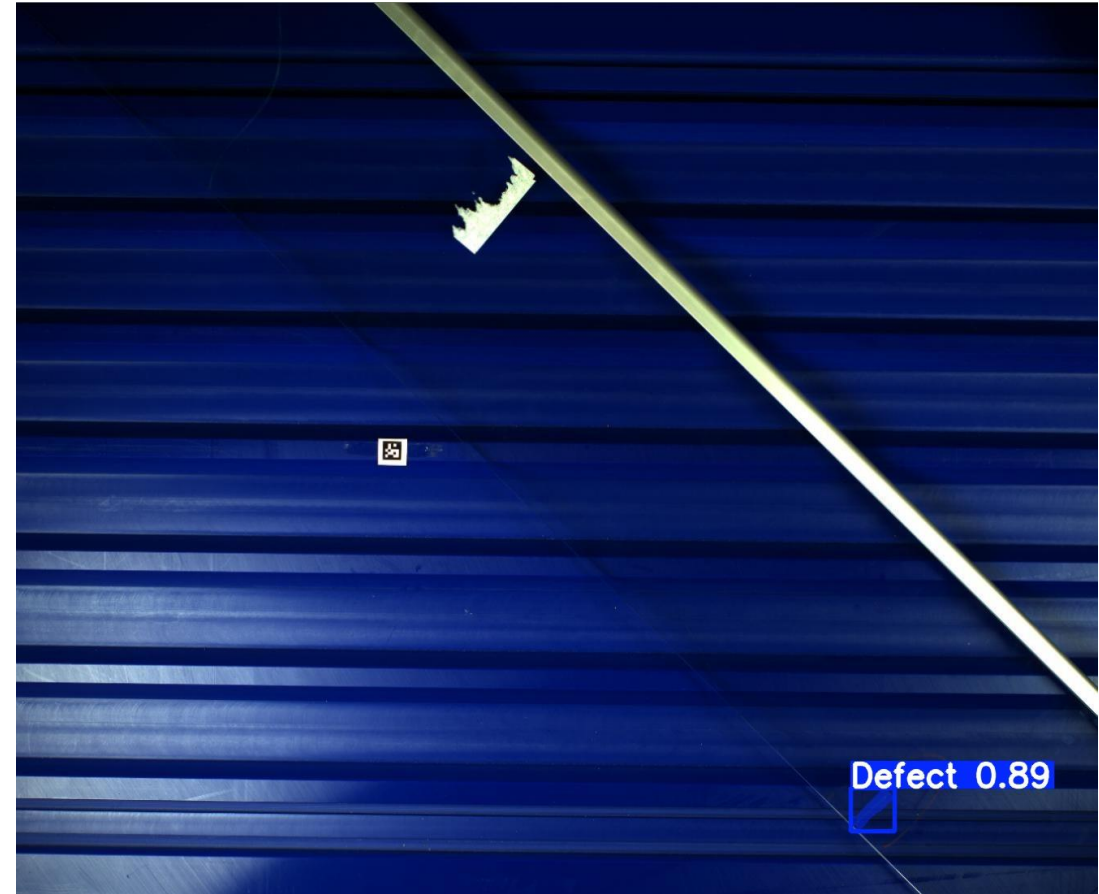


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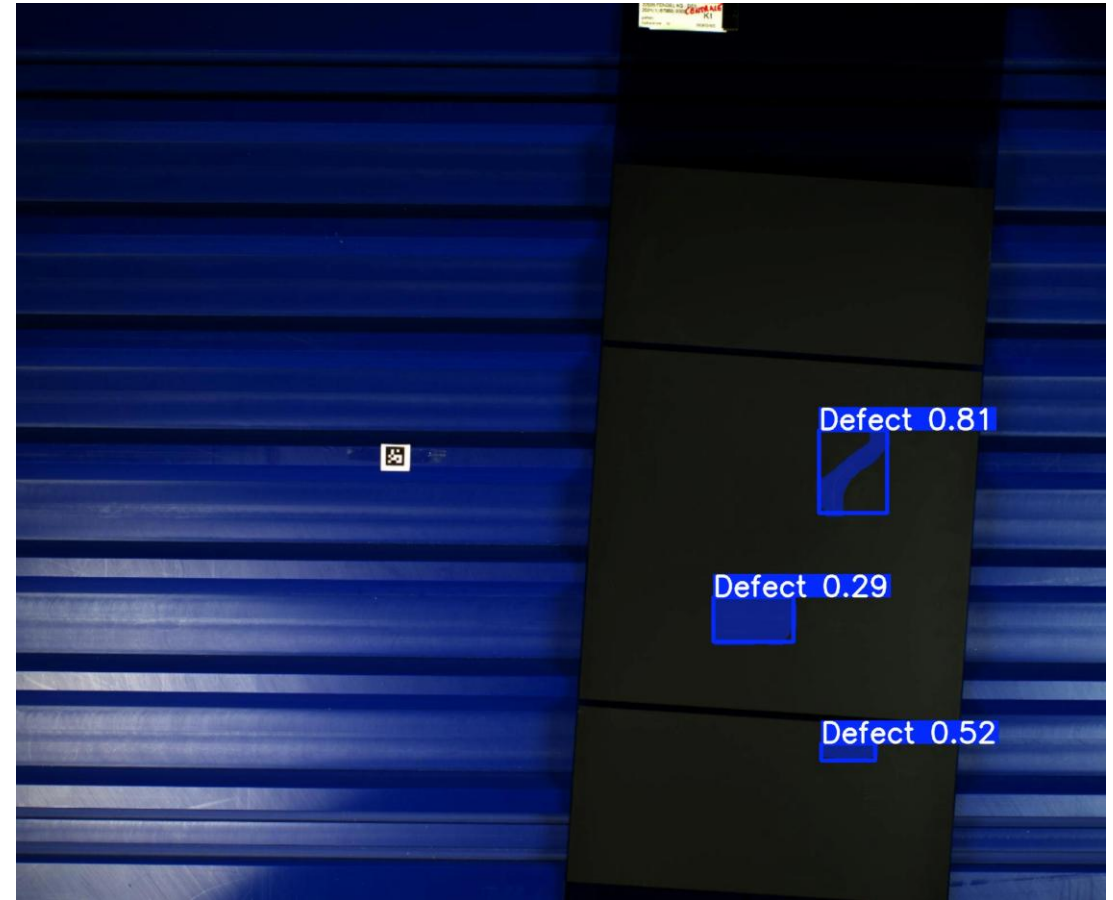


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Image source: Fraunhofer Italia

Benefits & Operator Feedback

Operators reported:

1. **Reduced cognitive load** and more consistent processes.
2. Error-prone manual transcription replaced by **automated QR decoding**.
3. **System is intuitive** → minimal training effort required.
4. Cost ~€20,000 → **affordable** for SMEs.
5. **Suitable beyond packaging**: furniture, electronics, mechanical parts.

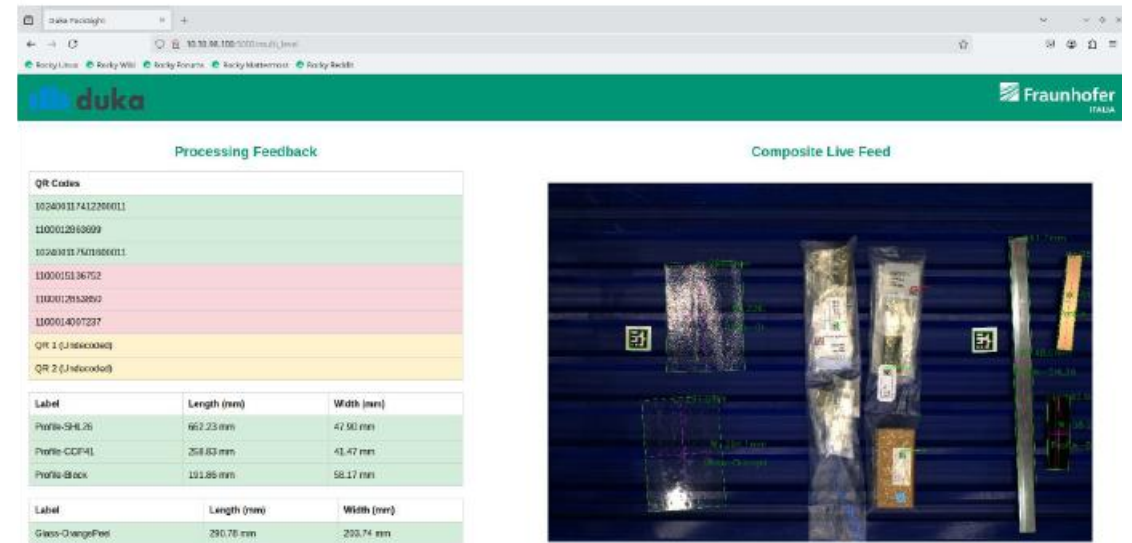


Image source: Fraunhofer Italia

Limitations and Future Directions

Some limitations:

- **Improve defect detection accuracy.**
- **Explore adaptive/self-learning lighting systems** for full autonomy.
- **Long-term validation** in real industrial environments under different environmental conditions.
- **Integration with** cyber-physical production systems (**CPPS**).
- **Expansion to multiple industrial domains** beyond packaging.

Let's get in touch for future collaborations!

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Thank you for your attention!
